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Homework for Chap 9

In a piece of semiconductor of GaAs, given the static dielectric constant is 12.9, the effective mass for electron is $0.067m_0$, the effective masses for heavy hole and light hole are $0.45m_0$ and $0.082\ m_0$, respectively. Here m_0 is the rest mass of electron, and $m_0=9.11\times10^{-31}$kg. Calculate the Bohr radius and binding energy of exciton ground state in GaAs. Please note that electron can form exciton with both heavy hole and light hole.

$$\mu = \frac{m_e m_h}{m_e + m_h}$$

$$\Rightarrow \begin{cases} \mu_{heavy} = 0.0583\ m_0 \\ \mu_{light} = 0.0369\ m_0 \end{cases}$$

$$E_B = E_g = 13.6eV \cdot \frac{\mu}{m_0} \cdot \frac{1}{\varepsilon^2} \quad \text{(ground state)}$$

$$\Rightarrow E_{B(light)} = 0.003eV$$

$$E_{B(heavy)} = 0.0048eV$$

Bohr Radius

$$a^* = a_0 \cdot \varepsilon \cdot \frac{m_0}{\mu}, \quad a_B^H = 5.29\times10^{-11}m$$

$$\Rightarrow a^*_{heavy} = 5.29\times10^{-11}\times12.9\times\frac{m_0}{0.0369m_0} = 1.85\times10^{-8}m = 18.5nm$$

$$a^*_{light} = 5.29\times10^{-11}\times12.9\times\frac{m_0}{0.0589m} = 1.17\times10^{-8}m = 11.7nm$$